FIRST APPROACH : determine the mean of y and sigma_y^2 solving the two - dimensional equilibrium conditions

In[154]:=

```
W[n_, y_, σy_] =
 Sqrt[(ω^2 + σy^2) / (2 π)] Exp[- (ω^2 + σy^2) y^2 / 2] (a n + c (ω^2 + σy^2) ω^2 y)
 raíz cuadrada                exponencial
```

Out[154]=

$$\frac{e^{\frac{1}{2} y^2 \left(-\sigma y^2-\omega^2\right)} \sqrt{\sigma y^2+\omega^2} \left(a\, n+c\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)\right)}{\sqrt{2 \pi}}$$

(A) We look for the equilibrium and average over n later (Assume n positive or negative)

In[155]:=

```
{D[Log[W[n, y, σy]], y], D[Log[W[n, y, σy]], σy]} // FullSimplify
 ⋯ logaritmo              ⋯ logaritmo                simplifica completamente
```

Out[155]=

$$\left\{-\frac{\left(\sigma y^2+\omega^2\right) \left(a\, n\, y+c\, \omega^2 \left(-1+y^2 \left(\sigma y^2+\omega^2\right)\right)\right)}{a\, n+c\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)},\ \sigma y \left(-y^2+\frac{1}{\sigma y^2+\omega^2}+\frac{2\, c\, y\, \omega^2}{a\, n+c\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)}\right)\right\}$$

In[159]:=

$$-\frac{\left(\sigma y^2+\omega^2\right) \left(a\, n\, y+c\, \omega^2 \left(-1+y^2 \left(\sigma y^2+\omega^2\right)\right)\right)}{a\, n+c\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)}-\left(-\left(\sigma y^2+\omega^2\right) y+\frac{c\, \omega^2 \left(\sigma y^2+\omega^2\right)}{a\, n+c\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)}\right) \text{ // FullSimplify}$$

simplifica completamente

Out[159]=

0

In[157]:=

$$\text{sy2} = \text{Solve}\left[a\, n\, y+c\, \omega^2 \left(-1+y^2 \left(s2+\omega^2\right)\right) == 0,\ s2\right] \text{ // Flatten}$$

resuelve aplana

Out[157]=

$$\left\{s2 \to \frac{-a\, n\, y+c\, \omega^2-c\, y^2\, \omega^4}{c\, y^2\, \omega^2}\right\}$$

In[163]:=

$$-y^2+\frac{1}{s2+\omega^2}+\frac{2\, c\, y\, \omega^2}{a\, n+c\, y\, \omega^2 \left(s2+\omega^2\right)} \text{ /. sy2 // FullSimplify}$$

simplifica completamente

Out[163]=

$$y^2 \left(2+\frac{a\, n\, y}{-a\, n\, y+c\, \omega^2}\right)$$

In[166]:=

$$\text{sy} = \text{Solve}\left[2 + \frac{a\,n\,y}{-a\,n\,y + c\,\omega^2} == 0,\ y\right] \text{// Flatten}$$

resuelve aplana

Out[166]=

$$\left\{y \to \frac{2\,c\,\omega^2}{a\,n}\right\}$$

In[167]:=

$$\frac{-a\,n\,y + c\,\omega^2 - c\,y^2\,\omega^4}{c\,y^2\,\omega^2}\ \text{/. sy // FullSimplify}$$

simplifica completamente

Out[167]=

$$-\frac{a^2\,n^2}{4\,c^2\,\omega^4} - \omega^2$$

Check that the solution is ok

comprueba

In[172]:=

$$\left\{-\frac{\left(s2 + \omega^2\right)\left(a\,n\,y + c\,\omega^2\left(-1 + y^2\left(s2 + \omega^2\right)\right)\right)}{a\,n + c\,y\,\omega^2\left(s2 + \omega^2\right)},\ -y^2 + \frac{1}{s2 + \omega^2} + \frac{2\,c\,y\,\omega^2}{a\,n + c\,y\,\omega^2\left(s2 + \omega^2\right)}\right\}\ \text{/. sy2 /.}$$

$$\text{sy // FullSimplify}$$

simplifica completamente

Out[172]=

$\{0, 0\}$

Direct computation

In[174]:=

$$\text{Solve}\left[\left\{-\left(\sigma 2 + \omega^2\right) y + \frac{c\left(\sigma 2 + \omega^2\right)\omega^2}{a\,n + c\,y\,\omega^2\left(\sigma 2 + \omega^2\right)} == 0,\right.\right.$$

resuelve

$$\left.\left. -y^2/2 + \frac{1}{2\left(\sigma 2 + \omega^2\right)} + \frac{c\,y\,\omega^2}{a\,n + c\,y\,\omega^2\left(\sigma 2 + \omega^2\right)} == 0\right\},\ \{y, \sigma 2\}\right] \text{// Flatten}$$

aplana

Out[174]=

$$\left\{y \to \frac{2\,c\,\omega^2}{a\,n},\ \sigma 2 \to \frac{-a^2\,n^2 - 4\,c^2\,\omega^6}{4\,c^2\,\omega^4}\right\}$$

There is a problem with then mean of y as function of n because the 1 / n behavior makes it non integrable about n = 0
Indeed, the integrand is odd so the principal value of the integral would be zero if we average over the real line

In[168]:=

```
Assuming[σn > 0 && ω^2 > 0 && σy > 0,
asumiendo
  Integrate[2 c ω^2/(a n) / Sqrt[2 π σn^2] Exp[-n^2 / (2 σn^2)], {n, -∞, ∞}]]
  integra                  raíz cuadrada     exponencial
```

Integrate: Integral of $\frac{c\, e^{-\frac{n^2}{2\sigma n^2}} \sqrt{\frac{2}{\pi}}\, \omega^2}{a\, n\, \sigma n}$ does not converge on $\{-\infty, \infty\}$.

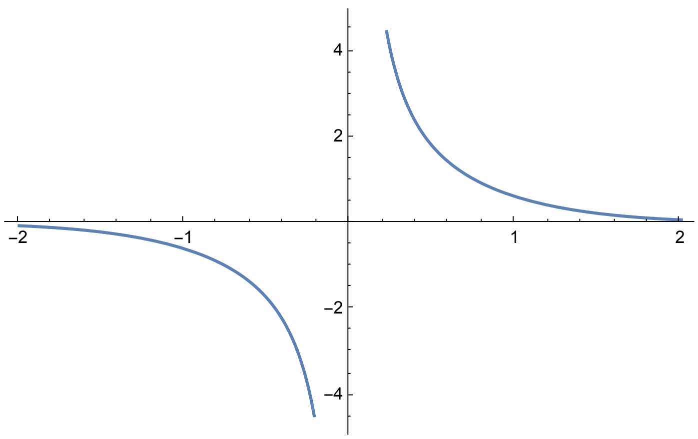
Out[168]=

$$\int_{-\infty}^{\infty} \frac{c\, e^{-\frac{n^2}{2\sigma n^2}} \sqrt{\frac{2}{\pi}}\, \omega^2}{a\, n\, \sqrt{\sigma n^2}}\, dn$$

In[179]:=

```
Plot[1 / n Exp[-n^2 / 2], {n, -2, 2}]
representa··· exponencial
```

Out[179]=



Moreover, the variance of y would be negative after averaging over n

In[169]:=

```
Assuming[σn > 0 && ω^2 > 0 && σy > 0,
asumiendo
  Integrate[(-a^2 n^2/(4 c^2 ω^4) - ω^2) / Sqrt[2 π σn^2] Exp[-n^2 / (2 σn^2)], {n, -∞, ∞}]]
  integra                                   raíz cuadrada     exponencial
```

Out[169]=

$$-\frac{a^2\, \sigma n^2}{4\, c^2\, \omega^4} - \omega^2$$

(B) What if n > n0, for n0 > 0

(we do not average over the whole real line thus avoiding the singularity at n = 0)

In[176]:=

```
Assuming[σn > 0, Integrate[Exp[-n^2 / (2 σn^2)], {n, n0, ∞}]] // FullSimplify
```
asumiendo integra exponencial simplifica completamente

Out[176]=

$$\sqrt{\frac{\pi}{2}}\ \sigma n\ \mathrm{Erfc}\left[\frac{n0}{\sqrt{2}\ \sigma n}\right]$$

In[177]:=

```
Assuming[σn > 0 && ω^2 > 0 && σy > 0 && n0 > 0,
  Integrate[(2 c ω^2)/(a n) Exp[-n^2 / (2 σn^2)], {n, n0, ∞}]]
```
asumiendo integra exponencial

Out[177]=

$$\frac{c\ \omega^2\ \mathrm{Gamma}\left[0, \frac{n0^2}{2\ \sigma n^2}\right]}{a}$$

So the average over y would be

In[178]:=

```
bary = (c ω^2 Gamma[0, n0^2/(2 σn^2)])/a / (Sqrt[π/2] σn Erfc[n0/(Sqrt[2] σn)]) // FullSimplify
```
función error complem··· simplifica completa

Out[178]=

$$\frac{c\ \sqrt{\frac{2}{\pi}}\ \omega^2\ \mathrm{Gamma}\left[0, \frac{n0^2}{2\ \sigma n^2}\right]}{a\ \sigma n\ \mathrm{Erfc}\left[\frac{n0}{\sqrt{2}\ \sigma n}\right]}$$

Does this have a good behavior in the limit n0 → 0

In[184]:=

```
Assuming[σn > 0 && n0 > 0, Series[Gamma[0, n0^2/(2 σn^2)] / Erfc[n0/(Sqrt[2] σn)], {n0, 0, 1}]] // Normal // FullSimplify
```
asumiendo serie normal simplifica completa

Out[184]=

$$-\frac{\left(n0\ \sqrt{\frac{2}{\pi}} + \sigma n\right)\left(\mathrm{EulerGamma} + 2\ \mathrm{Log}[n0] + \mathrm{Log}\left[\frac{1}{2\ \sigma n^2}\right]\right)}{\sigma n}$$

It diverges logarithmically as n0 → 0

Let us check whether the variance is positive or not if we integrate over (n0, ∞)

In[185]:=

$$\text{Assuming}\left[\sigma n > 0 \;\&\&\; \omega\text{^}2 > 0 \;\&\&\; \sigma y > 0,\right.$$

asumiendo

$$\left.\text{Integrate}\left[\left(-\frac{a^2 n^2}{4 c^2 \omega^4} - \omega^2\right) \text{Exp}[-n\text{^}2/(2\,\sigma n\text{^}2)],\ \{n,\ n0,\ \infty\}\right]\right]$$

integra exponencial

Out[185]=

$$\frac{\sigma n \left(-2 a^2 e^{-\frac{n0^2}{2\,\sigma n^2}}\, n0\, \sigma n - \sqrt{2\pi}\left(a^2 \sigma n^2 + 4 c^2 \omega^6\right) \text{Erfc}\left[\frac{n0}{\sqrt{2}\,\sigma n}\right]\right)}{8 c^2 \omega^4}$$

The variance would be

In[186]:=

$$\frac{\sigma n \left(-2 a^2 e^{-\frac{n0^2}{2\,\sigma n^2}}\, n0\, \sigma n - \sqrt{2\pi}\left(a^2 \sigma n^2 + 4 c^2 \omega^6\right) \text{Erfc}\left[\frac{n0}{\sqrt{2}\,\sigma n}\right]\right)}{8 c^2 \omega^4} \Big/$$

$$\left(\sqrt{\frac{\pi}{2}}\, \sigma n\, \text{Erfc}\left[\frac{n0}{\sqrt{2}\,\sigma n}\right]\right) \text{ // FullSimplify}$$

función error complem··· simplifica completamente

Out[186]=

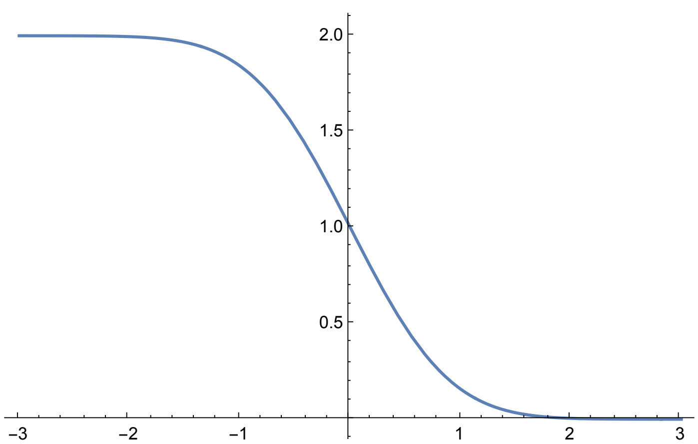
$$-\frac{\frac{a^2 \sigma n^2}{c^2} + 4\,\omega^6 + \frac{a^2 e^{-\frac{n0^2}{2\,\sigma n^2}}\, n0\, \sqrt{\frac{2}{\pi}}\, \sigma n}{c^2\, \text{Erfc}\left[\frac{n0}{\sqrt{2}\,\sigma n}\right]}}{4\,\omega^4}$$

In[221]:=

```
Plot[Erfc[x], {x, -3, 3}]
```

repr··· función error complementaria

Out[221]=



The variance we get is still negative because
the complementary error function is always positive

SECOND APPROACH : determine the mean of y and regard sigma_y ^ 2 as a parameter

In[188]:=

```
sy = Solve[-(σ2 + ω^2) y + c (σ2 + ω^2) ω^2 / (a n + c y ω^2 (σ2 + ω^2)) == 0, y] // FullSimplify
     resuelve                                                                  simplifica completa
```

Out[188]=

$$\left\{\left\{y \to \frac{-a\,n + \sqrt{a^2 n^2 + 4 c^2 \omega^4 (\sigma 2 + \omega^2)}}{2\,c\,\omega^2 (\sigma 2 + \omega^2)}\right\}, \left\{y \to -\frac{a\,n + \sqrt{a^2 n^2 + 4 c^2 \omega^4 (\sigma 2 + \omega^2)}}{2\,c\,\omega^2 (\sigma 2 + \omega^2)}\right\}\right\}$$

We check whether fitness is positive

In[143]:=

```
a n + c y ω^2 (σ2 + ω^2) /. sy // FullSimplify
                                   simplifica completamente
```

Out[143]=

$$\left\{\frac{1}{2}\left(a\,n + \sqrt{a^2 n^2 + 4 c^2 \omega^4 (\sigma 2 + \omega^2)}\right), \frac{1}{2}\left(a\,n - \sqrt{a^2 n^2 + 4 c^2 \omega^4 (\sigma 2 + \omega^2)}\right)\right\}$$

This means that we have to choose the first solution for bar y

(A) We compute the average over n (over the real line)

In[216]:=

```
Assuming[σn > 0 && ω^2 > 0 && σ2 > 0 &&
asumiendo
  Element[a, Reals] && Element[c, Reals] && σ2 + ω^2 > 0, Integrate[
  pertenece a  números r···  pertenece a  números reales    integra
  ((-a n + Sqrt[a^2 n^2 + 4 c^2 ω^4 (σ2 + ω^2)]) / (2 c ω^2 (σ2 + ω^2))) / Sqrt[2 π σn^2] Exp[-n^2 / (2 σn^2)], {n, -∞, ∞}]]
                                                                              raíz cuadrada   exponencial
```

Out[216]=

$$\frac{\sigma n\,\mathrm{Abs}[a]\,\mathrm{HypergeometricU}\left[-\frac{1}{2}, 0, \frac{2 c^2 \omega^4 (\sigma 2+\omega^2)}{a^2 \sigma n^2}\right]}{\sqrt{2}\,c\,\omega^2 (\sigma 2 + \omega^2)}$$

In[211]:=

```
bary1 = σn Abs[a] HypergeometricU[-1/2, 0, 2 c^2 ω^4 (σ2+ω^2) / (a^2 σn^2)] / (Sqrt[2] c ω^2 (σ2 + ω^2)) // FullSimplify
                                                                                                    simplifica completa
```

Out[211]=

$$\frac{\sigma n\,\mathrm{Abs}[a]\,\mathrm{HypergeometricU}\left[-\frac{1}{2}, 0, \frac{2 c^2 \omega^4 (\sigma 2+\omega^2)}{a^2 \sigma n^2}\right]}{\sqrt{2}\,c\,\omega^2 (\sigma 2 + \omega^2)}$$

Define the parameter ξ (it has dimensions of the inverse of the breeding rate)

In[212]:=

```
ξ = √2 c ω^2 (σ2 + ω^2) / (Abs[a] σn)
```

valor absoluto

Out[212]=

$$\frac{\sqrt{2}\ c\ \omega^2\ (\sigma 2 + \omega^2)}{\sigma n\ \text{Abs}[a]}$$

In[213]:=

$$\frac{\text{HypergeometricU}\left[-\frac{1}{2},\ 0,\ \xi\,\hat{}\,2\right]}{\xi}$$

Out[213]=

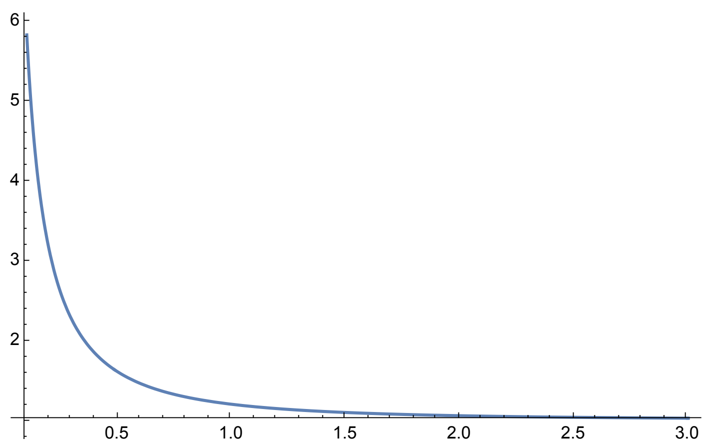
$$\frac{\sigma n\ \text{Abs}[a]\ \text{HypergeometricU}\left[-\frac{1}{2},\ 0,\ \frac{2\ c^2\ \omega^4\ (\sigma 2+\omega^2)^2}{\sigma n^2\ \text{Abs}[a]^2}\right]}{\sqrt{2}\ c\ \omega^2\ (\sigma 2 + \omega^2)}$$

In[214]:=

$$\text{Plot}\left[\frac{\text{HypergeometricU}\left[-\frac{1}{2},\ 0,\ \xi\,\hat{}\,2\right]}{\xi},\ \{\xi,\ 0.1,\ 3\},\ \text{PlotRange} \to \text{All}\right]$$

representación gráfica · rango de repr··· · todo

Out[214]=

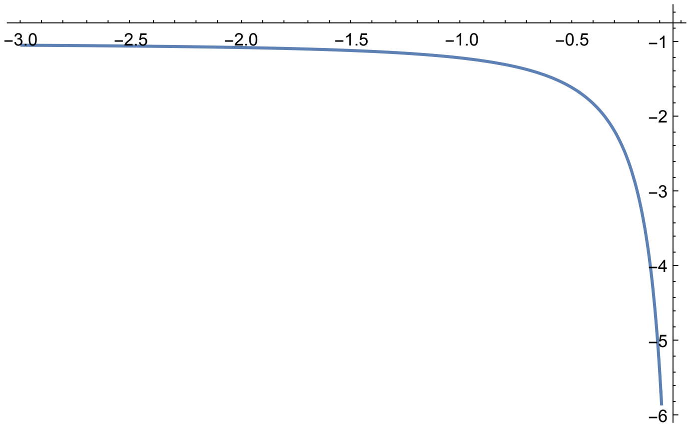


In[215]:=

```
Plot[HypergeometricU[-1/2, 0, ξ^2]/ξ, {ξ, -3, -0.1}, PlotRange → All]
```

representación gráfica · rango de repr··· · todo

Out[215]=



Expressed in terms of ξ, the average breeding rate is given in terms of the Tricomi confluent hypergeometric function

In[196]:=

```
bary2 = HypergeometricU[-1/2, 0, ξ^2]/ξ
```

Out[196]=

$$\frac{a\,\sigma n\,\mathrm{HypergeometricU}\left[-\frac{1}{2}, 0, \frac{2\,c^2\,\omega^4\,(\sigma 2+\omega^2)^2}{a^2\,\sigma n^2}\right]}{\sqrt{2}\,c\,\omega^2\,(\sigma 2+\omega^2)}$$

It diverges to $+\infty$ for $\xi \to 0$ from the right

In[206]:=

```
Series[HypergeometricU[-1/2, 0, x^2]/x, {x, 0, 1}] // Normal
```

serie · normal

Out[206]=

$$\frac{1}{\sqrt{\pi}\,x}$$

This expansion has the correct units because ξ has dimensions of the inverse breeding date

The average breeding date tends to one as $\xi \to +\infty$

In[208]:=

```
Series[HypergeometricU[-1/2, 0, x^2]/x, {x, ∞, 2}] // Normal
```

serie · normal

Out[208]=

$$1 + \frac{1}{4\,x^2}$$

It makes no sense that the average breeding date is negative (so probably this means that c positive has to be chosen)

(B) We check the maximum condition

In[219]:=

```
D[-(σ2 + ω^2) y + c (σ2 + ω^2) ω^2/(a n + c y ω^2 (σ2 + ω^2)), y] /. sy // FullSimplify
```
deriva | simplifica completamente

Out[219]=

$$\left\{-2\,\sigma 2-2\,\omega^2+\frac{a\,n\left(-a\,n+\sqrt{a^2\,n^2+4\,c^2\,\omega^4\left(\sigma 2+\omega^2\right)}\right)}{2\,c^2\,\omega^4},\ -2\,\sigma 2-2\,\omega^2-\frac{a\,n\left(a\,n+\sqrt{a^2\,n^2+4\,c^2\,\omega^4\left(\sigma 2+\omega^2\right)}\right)}{2\,c^2\,\omega^4}\right\}$$

In[220]:=

```
Assuming[σn > 0 && ω^2 > 0 && σ2 > 0 && Element[a, Reals] && Element[c, Reals] && σ2 + ω^2 > 0,
  Integrate[(-2 σ2 - 2 ω^2 + a n (-a n + Sqrt[a^2 n^2 + 4 c^2 ω^4 (σ2 + ω^2)])/(2 c^2 ω^4)) / Sqrt[2 π σn^2]
    Exp[-n^2 / (2 σn^2)], {n, -∞, ∞}]]
```
asumiendo | pertenece a | números r··· | pertenece a | números reales | integra | raíz cuadrada | exponencial

Out[220]=

$$-2\,\sigma 2-\frac{a^2\,\sigma n^2}{2\,c^2\,\omega^4}-2\,\omega^2$$

The second derivative averaged over n is negative at the critical point, so we have a maximum

(C) Compute the correlation between fitness and breeding date
constante

In[224]:=

```
ysol = y → (-a n + Sqrt[a^2 n^2 + 4 c^2 ω^4 (σy^2 + ω^2)])/(2 c ω^2 (σy^2 + ω^2))
```

Out[224]=

$$y\to\frac{-a\,n+\sqrt{a^2\,n^2+4\,c^2\,\omega^4\left(\sigma y^2+\omega^2\right)}}{2\,c\,\omega^2\left(\sigma y^2+\omega^2\right)}$$

We average the product of fitness and y to compute the correlation

In[227]:=

```
W[n, y, σy] y /. ysol // FullSimplify
```

simplifica completamente

Out[227]=

$$\frac{c\, e^{-\frac{\left(-a\, n+\sqrt{a^2 n^2+4 c^2 \omega^4 \left(\sigma y^2+\omega^2\right)}\right)^2}{8 c^2 \omega^4 \left(\sigma y^2+\omega^2\right)}}\, \omega^2 \sqrt{\sigma y^2+\omega^2}}{\sqrt{2\pi}}$$

In[231]:=

```
Assuming[σn > 0 && ω^2 > 0 && σy > 0 && Element[a, Reals] && Element[c, Reals],
  Integrate[(c e^(-(-a n + Sqrt[a^2 n^2 + 4 c^2 ω^4 (σy^2 + ω^2)])^2/(8 c^2 ω^4 (σy^2 + ω^2))) ω^2 Sqrt[σy^2 + ω^2] / Sqrt[2 π]) / Sqrt[2 π σn^2]
    Exp[-n^2/(2 σn^2)], {n, -∞, ∞}]]
```

asumiendo | pertenece a | números r··· | pertenece a | números re | integra | raíz cuadrada | exponencial

Out[231]=

$$\int_{-\infty}^{\infty} \frac{c\, e^{-\frac{n^2}{2\sigma n^2}-\frac{\left(-a\, n+\sqrt{a^2 n^2+4 c^2 \omega^4 \left(\sigma y^2+\omega^2\right)}\right)^2}{8 c^2 \omega^4 \left(\sigma y^2+\omega^2\right)}}\, \omega^2 \sqrt{\sigma y^2+\omega^2}}{2\pi \sqrt{\sigma n^2}}\, dn$$

Mathematica does not provide a closed - form formula